

Prostaglandin in Human Breast Cancer: Evidence Suggesting That an Elevated Prostaglandin Production Is a Marker of High Metastatic Potential for Neoplastic Cells^{1, 2}

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ABSTRACT—Prostaglandin (PG) production by human breast cancers was investigated in 91 lesions selected so that the distribution of histologic type was similar to that of the general population of mammary carcinomas. With regard to the shape characteristics of the tumors, PG production was higher in lesions classified T1 and T2 than in lesions classified T3 and T4 (T-classification is based on extent of tumor as graded by the International Union Against Cancer), higher in tumors exhibiting a high cellularity than in lesions with a low tumor cell density, and higher in tumors in which the cells were still adherent to each other. A high PG production was associated with the presence of neoplastic cells in tumor lymphatic and blood vessels and in axillary lymph nodes. PG production by node metastases was always higher than that by the primary tumor sites. The analysis of the stroma reaction and the presence of edema and necrosis suggest that an active PG synthesis occurred in lesions in which the tumor cell-surrounding stroma presented characteristics of low resistance to invasive growth of cancer cells. With regard to histologic differentiation and histoprosthetic grade of lesions, PG production was elevated in carcinomas that retained a minute part of the acinoductal differentiation and in tumors with a moderate or high degree of cancer. A lesion containing a steroid receptor (SR) tended to produce less PG than did an SR-negative tumor. PG production increased slightly according to ages and times of menopause of the patients. PG production occurred early in the natural course of breast cancer and was elevated in tumors at a time when active tumor invasion proceeded. By contrast, PG production decreased later in the course of tumor development. These results indicated that elevated PG production can be used as a marker of high metastatic potential for neoplastic cells in breast cancer.—*JNCI* 64: 1061-1070, 1980.

Synthesis of PG occurs in a number of malignant tumors (1-5), including human breast tumors (4-6), and in many transformed cells (7, 8). Yet little is known about both the production and the biologic significance of PG in cancer. Our laboratory is involved in developing and further characterizing human tumor specimens removed at surgery for studies of the mechanisms involved in hormone dependence of cancer and for use as screening systems for additional factors of identification of tumors and further therapeutic guidance (9-12).

This study was designed to describe the basic features of PG production in a representative population of human breast tumors and to examine the properties of PG-producing cells. PG production in each lesion was related to shape characteristics (size and clinical extension, cellularity, and cellular adhesiveness), cellular surrounding (stroma reaction, edema, and necrosis), histologic features (histologic type, differentiation, and histoprosthetic grade), related tumor-host response

features (lymphocytic reaction, tumor cell embolism, and axillary lymph node metastases), and hormone status (SR contents in tumors and age and menopause status of patients).

MATERIALS AND METHODS

At surgery, tumor samples were divided into two representative parts and immediately immersed either in liquid nitrogen for biochemical investigations or in Bouin's liquid for pathologic examinations as previously described (9-12). During the last 3 years, 1,400 primary and metastatic breast cancers were routinely investigated in the Institut Paoli-Calmettes. For the study of PG synthetase activity of breast cancer, a representative population of carcinomas was constituted from tumors that had been stored in liquid nitrogen, so that the histopathologic profile of the present series was similar to that of the general population of breast cancer and consisted of a limited number of cases. To prevent analyzing a single carcinoma of a rare HT, we fixed a lower limit of 3 specimens for the number of cases in each tumor class of this series. Finally, the specimens consisted of 71 primary tumors, 13 local recurrences or subcutaneous metastases from patients other than the 71 whose primary tumor was selected, and 7 axillary lymph node metastases of primary lesions included in the 71 primary tumors of this series. In addition, 4 benign lesions were examined. The age and endocrine status of the patient and the clinical tumor stage were recorded.

ABBREVIATIONS USED: B:EA=benzene-ethyl acetate; ER=estrogen receptor(s); HD=histologic differentiation; HPG=histoprosthetic grade; HSP=high-speed pellet; HT=histologic type(s); PG=prostaglandin(s); PR=progesterone receptor(s); SR=steroid receptor(s); TCD=tumor cell density; UICC TNM=International Union Against Cancer classification system based on extent of primary tumor (T), condition of lymph nodes (N), and absence or presence of metastases (M).

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We referred to the UICC TNM classification of human breast tumors as a basis for classifying the clinical extension and size of these carcinomas (13). Each lesion was examined for ER and PR as previously described (9-12).

In these studies, histopathologic features of each specimen were obtained from a double-blind examination of routine slides consisting of four cross sections along the axis of a tumor and, when results were unclear or doubtful, examination of additive serial slides; results of biochemical investigations were not known at that time. The cellularity of the tumor was determined according to an arbitrary classification of carcinomas on the basis of the relative proportions of tumor epithelial and stromal components on an increasing scale of tumor cell density (TCD I-TCD III) as previously described (9-12). The reference parameter of PG production, milligrams of protein, was used regardless of the relative abundance of cancer cells and stroma; its significance is discussed elsewhere (9-12).

On the basis of acinoductal differentiation and the prognostic value of the lesions, HT were recorded accordingly (14): 1) The most commonly found HT were the well-differentiated, atypical, and polymorphic carcinomas whose prognostic value was determined according to the histoprosthetic grades of Bloom and Richardson's modification of the system devised by Patey and Scarff (15). Furthermore, histologic differentiation of these lesions was recorded as 0%, more or less than 50%, or 100%. Poorly differentiated carcinomas were separated as atypical plus 0% differentiation or less than 10% differentiation carcinomas. Neither ductal nor papillary forms were observed on any section of the former lesions, whereas at least one of these organized structures was detected on one or several slides of the latter lesions, which were separated from the carcinomas with less than 50% differentiation (13, 14). 2) Special HT included the lobular and intraductal in situ carcinomas and lobular, infiltrating, colloid, isolated cells and medullary carcinomas (13, 14). Because of their own prognostic value, special HT were not graded (11, 14). In the examination of pathologic slides, the presence of tumor cell embolism, edema, and necrosis in lesions was recorded as positive or negative. Tumor cell embolism was considered positive when detached tumor cells, single or in clusters, were found in lymphatics and/or blood vessels that were defined in this series as at least endothelial-lined dilated spaces (16-20). Because the slides were trichrome stained (hematein, eosin, and saffran), the stromal reaction was determined as fibrohyalin according to the presence of fibrosis and (plus or minus) dense collagen fibers or as fibrinoid, which refers to the detection of elastotic and/or amyloid substances and, to a much lesser extent, to the presence of eosinophilic necrosis (21). The stromal lymphomonocytic reaction was considered negative (-) if lymphocytes were not seen, moderate (+) if cell number was low, and strong (++ and +++) when the stroma was characterized by an abundance or profusion of lymphomonocytes. The cell adhesiveness

in lesions was estimated according to the presence of isolated tumor cells in the tumor itself, in an independent manner of the tumor cell embolism, and was graded on a similar scale.

Analysis of PG synthetase activity.—We have previously shown that human breast tumors mainly produce PGE₂ in microsomes, whereas PGF_{2α} results from a PGE₂ metabolism within the cytosol. Therefore, PG synthetase capacity was measured by radioimmunoassay of the PGE₂ production in an HSP of breast tumor homogenates (4). At surgery, human breast tumors were isolated from conjunctive tissue and fat before the tumors were rapidly washed in isotonic saline solution, quick frozen, and stored in liquid nitrogen as previously described (9). In a cold room at 2° C, frozen tissue was powdered with a Thermovac apparatus and homogenized with a Polytron P-10 apparatus in a 1:5 (g:ml) tissue:buffer volume of 0.1 M (2*n*-morpholino)ethane sulfonic acid (pH 7.4) containing 2 mM CaCl₂, 1 mM MgCl₂, 2% glycerol, and 1 mM monothio-glycerol (MES buffer). Homogenates were centrifuged for 15 minutes at 800×g. The supernatant was then spun at 150,000×g for 1 hour at 2° C. The pellet was suspended in MES buffer and centrifuged again under the same conditions. The resulting pellet was suspended in a small volume of MES buffer and represented the HSP fraction. The protein content was measured by the method of Lowry et al. (22). Of the HSP fraction diluted in MES buffer to 0.5 mg protein/ml, we incubated 0.2 ml for 10 minutes at 37° C in a shaking water bath with 0.8 ml of MES buffer containing 1.25 mM reduced glutathione, 1.25 mM epinephrine, and 1.25×10⁻⁶ M Na salt arachidonate. Incubations were terminated by addition of 0.2 ml of citric acid (1 M), and radioactive PGE₂ was added in trace amounts to evaluate procedural losses. PG materials were extracted twice with 10 ml freshly decanted ethyl ether. The organic extracts were dried under a nitrogen stream, lyophilized, and then submitted to silicic acid chromatography according to (23) and modified as follows: The dried extracts were taken up with 2 μl acetic acid two times in 1 ml of a mixture of B:EA (8:2, vol/vol), and quantitatively transferred to the top of one of the columns (5-ml disposable Corning glass pipettes fitted with silane glass wool and silicone-treated by Siliclad; Clay-Adams). Each column contained 0.5 g of activated silicic acid (Mallinckrodt; 100-mesh) in B:EA (8:2). The flow rate was regulated by nitrogen pressure to about 1 drop/3 seconds. To reduce the background, we eluted neutral lipids, cholesterol, and free fatty acids, but not the hydroxy fatty acids, with 5 ml of B:EA (8:2). This eluate was then discarded. PGA's were eluted from the column by 5 ml of B:EA (6:4), PGE's by 5 ml of B:EA:methanol (6:4:0.3), and PGF₂'s by 4 ml of B:EA:methanol (6:4:2). Radioimmunoassays were performed on these fractions according to (4, 24, 25). Anti-PGE₂ antiserum was from the Institut Pasteur, Paris, France.

PG production will be expressed as picomoles PGE₂ produced for 10 minutes per 50 mg of HSP protein.

The statistical method analysis was based on the one-way analysis of variance, the block's method, and the chi-square method.

RESULTS

PG production was investigated in 91 neoplastic lesions and 4 benign diseases of the breast that were selected for this study according to their histopathology. These lesions were representative of a population of breast cancers (table 1). Its distribution of HT did not differ from that of general populations of mammary carcinomas (10, 11, 13, 14). PG production in neoplastic tissue varied widely from lesion to lesion. The lowest and highest values measured were 7 and 7,665 pmol PGE₂/50 mg protein/10 minutes. In benign breast disease, PG production was also observed in intraductal, tubular, and giant fibroadenoma and in an encysted abscess. PG synthesis activity was 34, 653, 1,027, and 164 pmol, respectively. In the encysted abscess containing many lymphocytes and plasma cells, a low level of PG production correlated with the intracellular production of PG observed in lymphocytes during activation by phytohemagglutinin (26).

Size, Cellularity, and Cellular Adhesiveness of Tumor

The clinician's measurement of the clinical extension of the carcinoma according to the UICC TNM classification of the tumors was used for the examination of PG production in relation to size and clinical extension of lesions (13). There was an inverse relationship between PG production and the size of the lesion, provided results from T1+T2 lesions were cumulative ($P<0.001$) (table 2). The mean PG production by T1+T2 tumors was statistically higher than that of T3+T4 lesions ($P<0.001$). Despite the fact that the difference was not statistically significant between T1 and T2 lesions, the highest activity was detected in T2 tumors. Therefore, the forthcoming results will refer to human breast cancer as a general population that

TABLE 2.—Relationship between clinical extension of the primary lesion and results of PG production analysis of carcinoma

Specification	Clinical extension of primary tumor: ^a			
	T1	T2	T3	T4
No. of specimens	11	32	24	4
PG production ^b	1,022±481	1,197±275	919±316 (n=24) 638±154 ^c (n=23)	401±201
	1,150±236 (n=43)		594±134 (n=28) ^d	

^a Determined according to the UICC TNM classification of breast tumors. n=No. of specimens of gathered populations.

^b Results are expressed as pmol PGE₂/50 mg protein per 10 min and are means±SE.

^c After withdrawal of a specimen (7,351 pmol PGE₂/50 mg protein/10 min).

^d $P<0.001$ (T1+T2 vs. T3+T4).

basically consisted of two classes of tumors, i.e., T1+T2 and T3+T4 lesions.

To examine whether PG production was related to the number of tumor cells, the cellularity of the lesions was correlated to PG synthetase activity. Results presented in table 3 show that PG production paralleled the rise of the cellularity in the lesions ($P<0.005$). In tumors with a high TCD, this relationship depended on the sizes of the lesions. It was maintained for the small-size tumors, whereas PG production by T3+T4 lesions was statistically lower than that of T1+T2 ($P<0.01$). PG production by T1+T2 lesions was lower in TCD I, similar in TCD II, and higher in TCD III than PG production by T1+T2 lesions of the general population, respectively ($P<0.01$). PG production by TCD I and TCD II carcinomas and also TCD III T1+T2 lesions could be considered as resulting from an integration of a constant production by each cell of either the whole tumor or a selected clone of the tumor. By contrast, PG production dropped in T3+T4 tumors with high TCD. One could speculate that either this relationship could not be ascertained or another phenomenon entered the time-course evolution of the tumor from T1+T2 to T3+T4.

To examine the tumor cell adhesiveness in relation to PG, we investigated the presence of nonadherent neoplastic cells in the lesion. Table 4 shows that there was an inverse relationship between the presence of isolated cells in the lesions and PG production ($P<0.01$). The highest PG-synthesizing activity was detected in lesions containing no isolated cells, whereas PG production was lower in tumors containing a high percentage of nonadherent cells. With respect to the sizes of the lesions, this phenomenon was essentially due to a lower PG production by T1+T2 lesions. Similar findings were observed for T3+T4 lesions that presented an abundance of isolated cells.

These results support the idea that PG production is higher in small tumors and in tumors of limited clinical extension with a high cellular density, provided neoplastic cells are adherent to each other. In

TABLE 1.—Distribution of HT of breast carcinomas^a

HT ^b	Present population, %	IGR, %	IPC, %
Isolated cells	4.3	3.6	1.5
Intraductal	9.8	2.3	3.9
Well-differentiated	4.3	9.3	9.3
Polymorphic	40.6	44.2	44
Atypical	27.5	34.7	32.7
Lobular infiltrating	6.6	3.9	3.9
Medullary	3.2	1.2	3.2
Colloid	3.2	0.6	1

^a The 95 cases of this study were compared with our general population at the Paoli-Calmettes Institute (IPC, 672 cases) and a 10-yr series established at the G. Roussy Institute (IGR, >1,000 cases).

^b Each HT is expressed as a percentage of the corresponding population.

TABLE 3.—Relationship between the TCD of carcinomas and results of PG production analysis of tissue

Specification	TCD ^a					
	I		II		III	
No. of specimens	16		53		14	
PGE ₂ production ^b	446±148		1,110±216		1,417±369 ^c	
Clinical extension of primary lesions ^d	T1+T2	T3+T4	T1+T2	T3+T4	T1+T2	T3+T4
No. of specimens	10	6	34	19	7	7
PGE ₂ production ^b	466±195	414±243	1,165±266	1,011±378	2,122±584	712±292 ^e

^a TCD of the tumor was estimated by a comparison of the density of cancer cells with that of the stroma on an increasing scale of I-III.

^b Results are expressed as pmol PGE₂/50 mg protein/10 min and are means±SE.

^c $P<0.005$ (TCD I vs. TCD II vs. TCD III).

^d Determined according to the UICC TNM classification of breast tumors.

^e $P<0.01$ (in TCD III: T1+T2 vs. T3+T4).

addition, one can conclude that notably in T1+T2 tumors, PG production decreased concomitantly with cell adhesiveness.

Tumor Cell Embolism and Lymphocyte and Monocyte Reaction in Tumors

To examine whether PG could be related to penetration of lymphatic and blood vessels by malignant cells, the presence or absence of tumor cell embolism in a lesion was correlated to PG production. As shown in table 5, there was evidence that the presence of emboli in lesions was associated with a higher PG production ($P<0.001$). This elevated PG production was due mainly to T1+T2 tumors. When PG production by emboli containing T1+T2 lesions was compared with that of the general population, a statistical difference was found ($P<0.005$). In contrast, whatever the size of lesions, the absence of tumor cell embolism was associated with a low PG production. These results showed that PG production correlated with the presence of neoplastic cells within the local vessel network in the primary tumor. Thereafter, we wanted to correlate PG synthesis capacity of T1+T2 lesions with the incidence of metastases in the corresponding axillary lymph nodes. As shown in table 6, there was a direct relationship between PG production and incidence of axillary lymph node metastases in lesions of constant sizes ($P<0.001$). These results showed that a high PG

production was associated with the presence of neoplastic cells in both local lymphatics and axillary lymph nodes. Furthermore, results in table 6 show that whatever the HD, PG production in axillary lymph node metastases was higher than that in the primary tumor. The presence or absence of lymphocyte and monocyte reaction was examined with respect to PG synthesis capacity in lesions. There was no relationship between PG production and lymphocyte presence in the lesions (table 7). These results showed that a lymphocytic reaction as a whole could not be related to either a defect or an excess in PG production.

Stroma Reaction, Edema, and Necrosis

The type of stroma in lesions was related to PG production (table 8). An intense fibrohyaline reaction of stroma was associated with a lower PG production ($P<0.001$). Furthermore, this stroma change was independent of tumor size. In contrast, the presence or absence of fibrinoid reaction revealed no statistical difference in PG production. However, the relationship between tumor size and PG production was reversed if an elevated fibrinoid reaction was noted; i.e., T3+T4 tumors produced more PG than did T1+T2 tumors. The presence of edema in lesions could be correlated with PG production regardless of tumor size. PG-synthesizing capacity in tumors in which no edema was detected was not statistically different from that of

TABLE 4.—Relationship between the presence of isolated tumor cells in carcinomas and results of PG production analysis of tissue

Specification	Presence of isolated tumor cells ^a					
	-		+		++ and +++	
No. of specimens	34		28		23	
PGE ₂ production ^b	1,278±320		1,075±230		698±142 ^c	
Clinical extension of primary lesions ^d	T1+T2	T3+T4	T1+T2	T3+T4	T1+T2	T3+T4
No. of specimens	24	10	18	10	11	12
PGE ₂ production ^b	1,295±360	1,235±694 ^e	1,229±316	798±299	776±248	626±158
		555±159				

^a -, None; +, moderate; ++, strong; +++, intense.

^b Results are expressed as pmol PGE₂/50 mg protein/10 min and are means±SE.

^c $P<0.01$ (- vs. + vs. ++ and +++).

^d Determined according to the UICC TNM classification of breast tumors.

^e PG production is given for total group ($n=10$) and then after ($n=9$) withdrawal of 1 specimen with the highest PG production (7,335 pmol).

TABLE 5.—Relationship between occurrence of neoplastic cell embolisms in tumor and PG production by the tissue

Specification	Tumor cell embolism ^a			
	—		+	
No. of specimens	44		40	
PGE ₂ production ^b	532±79		1,529±292 ^c	
Clinical extension of primary lesions ^d	T1+T2	T3+T4	T1+T2	T3+T4
No. of specimens	26	15	18	11
PGE ₂ production ^b	565±107	446±127	2,343±564	888±270
Incidence of lymph node metastasis	29.9% (n=24) ^e	—	66.6% (n=15) ^e	—

^a —, Absence; +, presence.^b Results are expressed as pmol PGE₂/50 mg protein/10 min and are means±SE.^c P<0.001 (— vs. +).^d Determined according to the UICC TNM classification of breast tumors.^e The incidence of lymph node metastasis, i.e., presence of neoplastic cells in lymph nodes regardless of the No. of involved nodes, was determined from pathologic examination of the axillary lymphatic curage corresponding to the primary tumors of which PG production is reported herein. Incidence is expressed as percentage of *n* lesions in which axillary curage was performed. P<0.001 (— vs. +).

TABLE 6.—Comparison of PG production by carcinoma between the primary tumor site and the corresponding axillary lymph node metastasis

Patient No.	HT of primary lesion ^a	PGE ₂ production ^b	
		Primary tumor	Axillary lymph node metastasis
195	Polymorphic, <50% diff	108	600
766	Polymorphic, 0% diff	540	2,585
885	Atypical	54	354
926	Polymorphic, 0% diff	108	600
986	Polymorphic, <10% diff	752	2,968
1,026	Polymorphic, 0% diff	17	566
1,079	Polymorphic, <50% diff	1,378	7,558

^a Following their acinoductal differentiation (diff), HT of polymorphic carcinomas was classified as 0% (atypical) diff, <50% and >50% diff, and 100% diff (well differentiated). Poorly differentiated carcinomas, i.e., atypical structures evidencing some ducts, were graded as <10% diff and separated from the carcinomas with >50% diff.^b Results are expressed as pmol PGE₂/50 mg protein/10 min and are means±SE.

the general population. Whatever the size of lesions, PG synthesis capacity was equal in tumors in which necrosis was absent. Low PG production by 1) T1+T2 lesions showing a fibrinoid reaction and 2) tumors having an intense fibrohyaline reaction suggests that the walling-in of tumor cells by a dense conjunctive tissue was associated with a low PG-synthesizing activity. Alternatively, this analysis of stroma reaction, edema, and necrosis suggests that an active PG synthesis occurred in lesions with stroma surrounding cancer cells suggestive of low resistance to invasive growth of malignant cells.

HD and HPG

Attempting the characterization of PG-producing cells with regard to histopathology and degree of malignancy of lesions, we investigated PG production in relation to HT, HD, and HPG of tumors. Table 9 shows PG production results of the most commonly found HT with respect to HD. The synthesis capacity was apparently unrelated to HD. Yet lesions with less than 10% differentiation presented a higher PG synthesis capacity than did all other classes (P<0.005). Despite the fact that there was no statistical relation-

TABLE 7.—Relationship between lymphocytic infiltrate in carcinoma and PG production by tissue^a

Lymphocyte reaction ^b	No. of specimens	PG production ^c	Extension of lesion ^d	No. of specimens	PG production ^c	Tumor cell embolism ^e	No. of specimens	PG production ^b
—	16	1,054±249	T1+T2	9	1,406±375	—	8	500±122
			T3+T4	7	824±304	+	8	1,688±407
+	44	990±236	T1+T2	25	1,047±302	—	24	492±111
			T3+T4	19	916±384	+	20	1,609±499
++ and +++	25	1,140±283	T1+T2	19	1,334±357	—	11	761±171
			T3+T4	6	524±213	+	14	1,437±480

^a According to the lymphocyte and monocyte reaction in the lesion, PG production by tissue is further compared in an independent way to 1) the size of lesions and 2) the presence of neoplastic cell embolism in carcinomas.^b —, None; +, moderate; ++, strong; +++, intense.^c Results are expressed as pmol PGE₂/50 mg protein/10 min and are means±SE.^d The clinical extension of the primary lesion was determined according to the UICC TNM classification of breast tumors.^e —, Absence; +, presence.

TABLE 8.—Relationship between the stromal reaction, necrosis, and edema of carcinoma and results of PG production analysis of tissue

Specification	No. of specimens	PG production ^a	Extension of lesion ^b	No. of specimens	PG production ^a
Necrosis ^c					
—	54	1,010±196	T1+T2	31	1,059±252
			T3+T4	22	951±329
+	30	1,159±256	T1+T2	20	1,413±351
			T3+T4	10	700±289
Edema ^c					
—	54	996±201	T1+T2	35	1,018±232
			T3+T4	19	733±185
+	28	1,126±269	T1+T2	16	1,565±407 ^d
			T3+T4	12	584±167
Type of stroma					
Fibrinoid ^e					
—	59	1,189±216	T1+T2	35	1,377±289
			T3+T4	24	800±301
+ and +++	26	843±163	T1+T2	18	750±185
			T3+T4	8	1,053±333
Fibrohyalin ^f					
+	67	1,310±200	T1+T2	40	1,468±245
			T3+T4	27	919±279
++ and +++	13	431±135 ^g	T1+T2	10	474±150
	13		T3+T4	3	428±220

^a Results are expressed as pmol PGE₂/50 mg protein/10 min and are means±SE.

^b The clinical extension of the primary lesion was determined according to the UICC TNM classification of breast tumors.

^c —, Absence; +, presence.

^d $P < 0.01$ (T1+T2 vs. T3+T4).

^e —, None; + and ++, moderate or abundant.

^f +, Moderate; ++ and +++, abundant or profusion.

^g $P < 0.001$ (+ vs. ++ and +++).

TABLE 9.—Relationship of HT of carcinomas and results of PG production analysis of tissue^a

HT	No. of specimens	PG production ^b
Common ^c		
Atypic, polymorphic 0% diff	25	741±136
Polymorphic <10% diff	17	2,176±470 ^d
Polymorphic <50% diff	12	760±177
Polymorphic >50% diff	5	535±268
Well-differentiated, polymorphic 100% diff	7	444±347
		(n = 24)
Special		
Isolated cells	4	125±75
Colloid	3	231±136
Medullary	3	180±23
Intraductal	9	245±71
Lobular infiltrating	6	312±64
Well-differentiated	4	604±347
	3 ^e	267±19

^a According to the acinoductal differentiation and the prognostic value of lesions, HT of carcinomas were recorded as commonly found HT (atypical, polymorphic, and well-differentiated carcinomas) and special HT (lobular infiltrating, lobular, intraductal in situ, colloid, medullary, and isolated carcinomas).

^b Results are expressed as pmol PGE₂/50 mg protein/10 min and are means±SE.

^c Following their acinoductal differentiation (diff), polymorphic carcinomas were classified as 0% diff plus atypical, <50% or >50% diff, and 100% diff plus well-differentiated. Poorly differentiated carcinomas, i.e., atypical structures evidencing some ducts, were graded separately as 10% diff.

^d $P < 0.005$ (10% diff vs. other classes).

^e After withdrawal of a specimen (1,610 pmol PGE₂/50 mg protein/10 min).

ship, PG synthesis tended to slightly increase from well-differentiated lesions to carcinomas of less than 50% differentiation, to reach a maximum in tumors of 10% differentiation, and then to exhibit a drop in atypical lesions. With respect to HD, PG production was not related to tumor size. Particularly, in atypical tumors plus tumors with 0% differentiation, PG production was 677±145 ($n = 13$) and 867±229 ($n = 12$) pmol PGE₂ in T1+T2 and T3+T4 lesions, respectively. In lesions with special HT (table 9), PG synthesis capacity was detected at very low levels, and no difference was noted. In agreement with the results found by a comparison of PG production with cell adhesiveness, PG synthesis capacity of isolated cell carcinomas was very low. Medullary carcinomas were also low PG producers because these tumors are thought to be the site of intense immunologic host-tumor reaction (14, 16, 20).

Table 10 compares the PG production of the most commonly found HT with their HPG. PG production by HPG II carcinomas was higher than that in HPG I ($P < 0.005$) and HPG III carcinomas ($P < 0.05$). These results suggested that PG production was elevated in in situ carcinomas that retained a minute part of the initial acinoductal differentiation and in tumors with a moderate or high degree of malignancy.

SR Contents in Tumor: Age and Menopause in Patients

Having previously demonstrated that a positive SR content was associated with a moderately aggressive

TABLE 10.—Relationship of HPG of carcinomas^a and results of PG production analysis of tissue

Specification	HPG					
	I		II		III	
No. of specimens	9		23		27	
PG production ^b	620±215		1,874±406 ^c		1,037±210	
Clinical extension of primary lesions ^d	T1+T2	T3+T4	T1+T2	T3+T4	T1+T2	T3+T4
No. of specimens	9	—	16	7	14	12
PG production ^b	620±215	—	1,847±453	1,906±1,160	1,112±360	961±340

^a The carcinomas with a common HT were graded according to the method of Bloom and Richardson (15). Because of their own prognostic value, special HT were not graded.

^b Results are expressed as pmol PGE₂/50 mg protein/10 min and are means±SE.

^c $P<0.005$ (II vs. I), $P<0.05$ (III vs. II).

^d Determined according to the UICC TNM classification of breast tumors.

nature of breast carcinomas (10–12), we correlated the presence of ER and PR in lesions to PG production to examine the PG synthesis activity in relation to hormone dependence as well. Results in table 11 show that the mean PG production was higher in ER-negative than in ER-positive lesions, yet this difference was not statistically significant ($P<0.1$) in this series. Because we have previously demonstrated that the presence of PR is linked to that of ER (10–12), PG production was correlated to PR presence in ER-positive lesions. In PR-negative lesions, PG production was higher in axillary metastases than in the primary tumors ($P<0.02$). However, T1+T2 primary tumors containing both ER and PR produced more PG than did either T3+T4 tumors or axillary metastases, yet node metastases produced lower amounts of PG than did T3+T4 lesions. These differences were not statistically significant. According to the hormone status of patients, PG production was 896 ± 166 ($n=16$) and $1,068\pm210$ ($n=59$) pmol PGE₂ in premenopausal (25–50 yr old) and postmenopausal (50–90 yr old) patients, respectively. Also, there was a slight but statistically insignificant increase in PG production according to age and

menopausal status of the patients. When the relationship between tumor size and PG production was considered separately before and after menopause, no significant relationship was noted.

DISCUSSION

The parameters governing PG production in human breast cancer are progressively defined. PG production by human breast tumors confirmed by other workers was also observed in our present study (4–6, 27). Having previously demonstrated that PGE₂ production by the microsome fraction from human breast tumor reflects the PG synthesis capacity of a given tumor specimen (4), we quantified PG production and correlated it with the tumor characteristics at a given time during the natural course of the lesion, i.e., at the time of surgery.

Inasmuch as PG synthesis was elevated in human breast tumors as long as they remained relatively small, PG production probably occurs early in the natural course of tumor development. Thereafter, either PG-producing cells are no longer present in the primitive lesion or there is a decrease in the production of PG (whatever the cause, through production repression or cessation of stimulating influence). Because PG production paralleled the increase of TCD, regardless of any other factors, PG production conceivably reflects an integrative value of a constant production by cancer cells within a given tumor specimen.

Neoplasms presumably consist of both producing and nonproducing cells. A similar heterogeneity was observed concerning the presence of SR (10–12). Thus PG production reflects the synthesis by a variable percentage of cells. However, the fact that in TCD III carcinomas this relationship was not retained for the largest lesions suggests that PG lower as a tumor grows and overcomes the production rise which paralleled the cellularity. As observed with regard to lesion size, this argues for the early occurrence of an active PG synthesis.

The cellular surrounding with regard to other non-cellular factors, which reflects the concomitant presence of an elevated PG production and other tumor characteristics as well, is suggestive of an intense metastatic activity. A markedly reduced PG production

TABLE 11.—Relationship of SR contents of carcinoma and results of PG production analysis of tissue

Specification	No. of specimens		PG production ^a	Total
ER-negative ^b	32		1,107±283	
ER-positive ^b	61		861±121 ^c	
ER+, lesion	T1+T2 ^d	T3+T4	Axillary lymph node metastasis	
PR-, No. of specimens ^b	9	10	6	25
PR-, PG production ^b	539±200	579±154	1,330±489 ^c	745±157
PR+, No. of specimens ^b	22	9	5	36
PR+, PG production ^b	1,122±254	572±277	816±207	942±175

^a Results are expressed as pmol PGE₂/50 mg protein/10 min and are means±SE.

^b ER and PR were considered positive (+) or negative (–) regardless of receptor content.

^c $P<0.1$.

^d The clinical extension of the primary lesion was determined according to the UICC TNM classification of breast tumors.

^e $P<0.02$ (axillary lymph node metastasis vs. primary tumors).

was observed in lesions revealing an intense stromal reaction; only T1+T2 lesions are of concern for fibrioid-type stroma. The presence of elastosis in human breast cancer is a factor of good prognostic value and of low tumor activity because its presence was found to be associated with both SR and low HPG (16, 20, 21, 33). Alternatively, tissues rich in dense fibers are thought to be a successful barrier through the mechanical process of walling-in of the neoplastic cells and/or are capable of inhibiting tumor-induced vascular proliferation by the production of protease inhibitors (28-33). Whatever the mechanisms, the fact remains that the presence of stromal reactions suggestive of host-tumor interrelationships restricting tumor invasion was associated with a low PG production in lesions.

The loss of cell adhesiveness in the primary carcinomas was found to be associated with a lower PG production. The invasive properties of tumor cells have been attributed in part to their reduced adhesiveness (30), and it is well known that large tumors show a higher incidence of distant metastases (16, 17). Yet detachment and embolization of tumor cells are presumably continuous processes which begin immediately with the growth of the primary malignant tumors (30-35). This view is supported by the fact that surgical removal of the lesion, whatever its size, does not ensure survival of the patient. Furthermore, 25% of breast carcinomas less than 1 cm present metastases (17, 34). Also, one may consider that issuing of cancer cells and clusters from the parent tumor have previously occurred and still continue as the loss of cell adhesiveness appears in lesions.

The PG production by breast carcinomas was then correlated to local and regional tumor spread by examination of the occurrence of tumor cell embolisms and the incidence of axillary lymph node metastases. Particularly in T1+T2 lesions, the presence of tumor cell embolisms appears to be strongly related to an elevated PG production. Whatever the size of the lesion, a high PG production was found to be associated with the presence of detectable tumor cell embolisms and a markedly increased incidence of axillary lymph node metastases. In agreement with the relationship between elevated PG production and stromal reaction revealing a favorable tissue structure for tumor invasion, the high occurrence of tumor cell embolisms and the presence of node metastases in PG-producing lesions suggest that elevated PG production occurs simultaneously with active tumor spread. Moreover, the comparison of PG production by the primary tumor site and the corresponding axillary lymph node metastasis shows that PG production is always higher in metastases than in primary tumors. Also, transfer of PG-producing cells from the primary tumor to another site, i.e., a metastasis, could explain the lowering of PG production as both tumor growth and cell adhesiveness are reduced.

The relationship between PG production and HPG illustrates the assumption that PG synthesis is elevated while tumor invasion progresses, i.e., at a time of better prognosis, one that will lead to an advanced

disease status. From a morphologic and histologic standpoint, HPG reflects the degree of malignancy of the cancer with respect to survival rates of the patients (15). PG production was low in HPG I lesions and was not found to be more pronounced in lesions with the highest degrees of malignancy but in those with a moderate degree. As stated above, this does not correspond to advanced disease but to the development of metastases, i.e., the step leading inevitably to multiple secondary tumor growths—a worse prognosis.

Recently, Fidler and co-workers (30, 36) proposed that there may exist a highly metastatic variant cell that possesses the intrinsic characteristics necessary to assume the cascade of metastatic events before exhibiting secondary foci. It is important to identify these cells by the presence of a specific marker. These cells may have properties enabling them to evade host defenses that mainly involve immunologic defenses. The immunologic features that have been investigated in this study concern the lymphocyte and monocyte reaction which was found to be unrelated to PG synthesis activity. Because the pathologist is actually unable to distinguish different classes of lymphocytes on routine slides, particularly with respect to suppressor cells (37), the possibility still exists that more precise identification of the lymphocyte reaction would reveal a correlation with PG production. Yet there have been several suggestions that the production of PG by tumor cells might be a mechanism whereby host immune responses are somehow diffused or subverted (38-41). According to the stipulations of Fidler and co-workers (30, 36) that cells should possess the characteristics necessary to assume the metastatic events (30, 36), one should remember that tumor cell traits enabling invasion and metastases concern modifications in cell surface properties, adhesive capacities, cell locomotion, and enzyme secretion. Insofar as the survival and growth of these cells will be ensured with the use of host homeostatic mechanisms, most of these effects could be expected to result from the loss of the host-tissue component relationships of reciprocal dependence and from the improvement of cellular autonomy. We recently demonstrated that hormone deprivation leads to normal cell accumulation in the thyroid gland and release of PGE_2 in culture medium (25), whereas morphologic characteristics of the normal thyroid gland are not retained in these cells (42). However, transformation of normal cell lines is often followed by such an increase in PGE_2 synthesis. PG presence in growth medium resulted in sustained production and release of cyclic AMP that in turn restored normal growth and morphologic structure to transformed fibroblasts (7, 8, 43, 44). Information on the long-term regulation of PG synthesis is scarce, and the comparison between normal, transformed, and neoplastic cells is sometimes hazardous. Nevertheless, a role in maintenance of the physiologic steady state of these systems was given to PGE_2 , i.e., a relationship between PG production and the autonomy of cells (25). Contrariwise, we have demonstrated that the presence of SR, which are proposed to reflect hormone depen-

dence of human breast cancer, was related to 1) a high degree of acinoductal differentiation in lesions and 2) a low grade of malignancy of the cancer and rate of cell division (9-12). Hormone dependence was demonstrated to oppose cellular autonomy, and a progression from the former to the latter was shown to be accomplished by sequential clone selection in an orderly sequence (45). Also, conceivably the absence of SR reflects a loss of cellular factors responsible in part for mediation of growth-restricting or growth-stimulating and differentiation-restricting or differentiation-stimulating influences. Consequently, this absence of SR may be associated with the improvement of the autonomy of these cells insofar as it parallels a loss of the host-tissue relationships of reciprocal dependence that were effective in the physiologic status. We now report PG production to be elevated in SR-negative lesions and in poorly differentiated carcinomas. Concerning PG production increments with lower HD [this paper and (25)], a drop in PG production independent of size was observed in atypical lesions. As stated above, it is more likely that, similar to the lowering of the PG production following the evolution of tumors, cellular spreading of PG-producing cells in both atypical and HGP III tumors begins very early so that these cells do not accumulate at the primary site. Clinical reports support this hypothesis. In particular, patients with poorly differentiated lesions reveal a higher incidence of axillary node metastases (17). They also have lower survival and recurrence-free rates than do patients with differentiated lesions (17).

In conclusion, we present evidence that PG production occurs very early in the natural course of human breast cancer. An elevated PG production seems to be a factor that characterizes cancer cells capable of performing the cascade of events in metastases and evading host immune responses. Yet PG production by itself could not be considered a marker of neoplasia because we have shown that benign breast disease produces PG (46) as well as benign prostatic hyper trophy (47). These productions were within the PG synthesis range detected in breast cancer. By contrast, the findings reported here have encouraged us to attempt to develop an elevated PG production as a marker of high metastatic potential for neoplastic cells in breast cancer. In addition to histopathologic examinations and SR analysis, PG production measurement will perhaps allow a better staging of neoplastic lesions with respect to the temporal and prospective clinical extension of the disease and of the resulting prognosis for the patient. Yet most of the findings reported here have led to conclusions and/or hypotheses remaining to be confirmed by both in vitro experiments on cells in culture and follow-up of the patient's clinical course and survival.

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